

The Evaluator Is Part of the Forecast: A Forensic Audit of ECN-BENCH

Minh Hien Nguyen

Abstract—An LLM that converts a multi-agent transcript into a probability distribution is part of the forecasting system, rather than a neutral scoring function. We study this measurement boundary through a retrospective audit of ECN-BENCH, a social-simulation testbed with four model campaigns over 30 resolved events. On 132 simulated units with identical stored evidence, questions and options, near-uniform forecasts occur in 31 readings by the original extractor and one reading by a replacement. Repeated readings by the replacement also vary: the mean within-unit Brier range is 0.208–0.375 across campaigns. These observations establish sensitivity to the deployed extraction configuration; they do not identify a uniquely correct reading of the swarm. A source audit documents schema bypass and silent failure handling, but changing the evaluator also changes other configuration details. An existing truncation and enforcement experiment on a proxy model does not establish either factor as the cause of the original near-uniform outputs. Exploratory forecasting results are evaluated separately. On 20 events resolving after the campaign models’ release dates, the Brier contrast between the two simulation conditions is $+0.112[+0.030, +0.192]$ under averaged replacement-evaluator readings and $+0.081[-0.003, +0.170]$ under another evaluator, with event-cluster 95% bootstrap intervals. The latter includes zero. This subset is not protected against evaluator knowledge leakage; some injections target the wrong event, and the null payloads are not length-matched. An existing single-agent baseline does not establish a simulation advantage. The contribution is an auditable case study of extraction sensitivity, with explicit boundaries on what the available data can say about forecasting and deliberation.

Index Terms—LLM evaluation, probability extraction, multi-agent simulation, forecasting, measurement reliability, reproducibility, benchmark audit.

I. INTRODUCTION

A social-simulation forecast involves at least two distinct transformations: agents turn an event dossier into a discussion, and an extractor turns that discussion into a probability vector. When the extractor is itself an LLM, its model, prompt, serving environment and post-processing can affect the reported forecast. A change in forecasting score can consequently reflect a change in the discussion, a change in its interpretation, or both.

ECN-BENCH was originally built to investigate whether social simulation improves forecasting relative to direct elicitation. Auditing its execution records exposed a more immediate question: how stable is the measurement used to evaluate that claim? Near-uniform probability

vectors appeared frequently in the original extraction pass. Substituting the deployed extractor substantially changed those vectors on fixed evidence. The audit also uncovered a shared reference condition, failed simulations, catalogue collisions and incomplete telemetry.

This paper reports the resulting measurement study. Its primary observation is a comparison on fixed stored evidence; its secondary observations describe repeat variability and existing follow-up experiments. Forecasting contrasts are retained as exploratory measurements of the complete pipeline. They are not evidence that a swarm has acquired knowledge unavailable to its members, nor that 300-agent deliberation improves on a single model call.

The contributions are:

- 1) An artifact-verifiable comparison of two extraction configurations on 132 simulated units, with the evidence, question, options and outcome metadata checked for equality before pairing.
- 2) A reanalysis of repeated extraction, a proxy truncation/enforcement grid and a single-agent baseline, with failures and unequal measurement budgets made explicit.
- 3) A reproducible account of how execution defects, statistical dependence and retrospective knowledge exposure constrain a forecasting benchmark’s claims.

All analyses in this revision use existing artifacts. This is a retrospective case study, not a preregistered confirmatory experiment. The aim is to describe a consequential measurement failure mode without assigning an unsupported mechanism to it.

II. RELATED WORK

LLM-as-a-judge research already treats evaluators as imperfect measurement devices. Zheng et al. [2] document biases in LLM judgments; Haldar and Hockenmaier [3] directly investigate inconsistency across repeated judgments. We do not claim that repeated-judge instability is new. Our narrower contribution concerns the extraction of categorical probabilities from social-simulation evidence and the associated source-code and data-provenance audit.

Proper scoring rules provide a framework for evaluating probabilistic forecasts. Gneiting and Raftery [1] distinguish calibration from sharpness: sharpness concerns the concentration of the predictive distribution and does not depend on which outcome is eventually observed. A higher Brier score alone cannot locate an error in either the

agents’ reasoning or the extractor’s interpretation. Likewise, a sharp distribution is not necessarily a faithful or accurate one.

OASIS [4] supports large-scale LLM social simulations. ECN-BENCH uses the MiroFish-Offline pipeline built on that framework; the simulation infrastructure is the testbed for this audit, rather than a new simulation algorithm. ForecastBench [5] evaluates questions whose answers are unknown when forecasts are submitted. That prospective design addresses a limitation our retrospective release-date filter cannot resolve: every LLM component capable of generating a forecast must be considered when evaluating knowledge leakage.

III. TESTBED AND EVIDENCE PROVENANCE

A. Events and conditions

The analysed catalogue contains 30 resolved events curated from Polymarket, with categorical outcome sets of size $K = 2-9$. We retain the catalogue’s identifiers: 15 C events, seven S events and eight T events. These labels are record identifiers; we do not treat their historical topical grouping as a validated taxonomy. The catalogue’s descriptive header contains superseded counts, so the executable analysis validates the observed campaign units.

Each campaign contains 30 records for each of three conditions:

- **Condition A: shared direct reference.** The stored reference evidence is reused across campaigns. It is not a separately executed baseline for each campaign model.
- **Condition B: nominally relevant injection.** A configured 300-agent, 60-round simulation receives an event-specific update at round 30.
- **Condition C: null injection.** The same configured workflow receives an unrelated administrative update at round 30.

A unit is a campaign–event–condition record. Thus 360 unit records exist, but neither the four copies of the shared reference nor repeated evaluator readings constitute independent simulation replications. Agent count is a configuration parameter, not the number of independent human-like observations or a guarantee that every agent contributed substantive discussion.

The intended B/C manipulation was relevant versus unrelated information. The released inputs implement a less controlled comparison. Seven injection identifiers are aligned to a stale catalogue. Among the 20 post-release events used later, C6 concerns a papal conclave but receives a Fed update, T6 concerns the Irish election but receives a GPT-5 update, and T8 concerns the Canadian election but receives a Bitcoin update. Repairs corrected simulation questions while retaining this injection bank. Consequently B is described throughout as *nominally* relevant.

The bank also fails the intended length match: relevant-update bodies contain 1.51–2.37 times as many characters

as null bodies (median 2.04), and all null bodies use the same text. This is a measurement of the released input bank, not an estimate of the subsequent discussion length. Relevance, wording, length and stimulus diversity are therefore not separated by this design.

B. Campaign identity and recovery

The four campaign labels are Llama-3.1-8B, Mistral-7B, Qwen2.5-7B and Qwen2.5-14B. The Llama campaign originally used an AWQ-INT4 checkpoint under local vLLM; its recovery used a hosted Llama endpoint. Mistral used the v0.1 instruct model with local full-precision serving; Qwen2.5-7B used a local Ollama/GGUF deployment. The Qwen2.5-14B artifacts span original and recovery executions, and their names and serving descriptions are not consistent enough to establish one numerical-precision setting for the whole campaign. We retain the label without treating it as a controlled size or quantization comparison. Release generation, serving and precision are confounded across campaigns.

The Llama backend outage affected 23 events; catalogue correction brought the recovery set to 24 events. An initial recovery incorrectly constructed empty evaluator evidence and was superseded. The final stored recovery records contain rebuilt evidence and replacement scoring. Only the six unaffected events (C3, S2–S6) can enter the comparison with the original July extractor.

The full question-level audit additionally finds Qwen2.5-14B T1 still asks about a Fed decision rather than the UAW question carried by the other campaigns. Its options and outcome happen to match, so vector validation does not detect the topic collision. T1 is outside the primary 20-event subset; it is excluded across all campaigns in the extended 29-event question-aligned summaries, while its stored rows remain visible.

Qwen2.5-14B required six wrong-topic events to be re-simulated before the present revision. A separate unit, C9 in Condition C, retains an outage-like telemetry signature. Its scored probabilities survive, but its simulation integrity is uncertain. We report full-set descriptive results and a sensitivity analysis excluding the entire C9 event across campaigns, alongside the wrong-injection events, to keep a balanced comparison.

C. Extractor identities and read budgets

The archive separates evaluator files:

- **Evaluator J:** the original local DeepSeek-R1-Distill-Qwen-14B deployment, identified in the pipeline as `deepseek-r1-14b`.
- **Evaluator A:** the recorded hosted identifier `deepseek/deepseek-v4-flash-0731`. Three campaigns have an initial pass and three additional reads. Qwen2.5-14B has three reads only.
- **Evaluator C:** the recorded hosted identifier `openai/gpt-5.6-luna-pro`, with one read per unit.

These hosted names are execution identifiers from the archive, not evidence that the same backend or weights remain available. Provider routing and backend determinism were not independently verified.

The file `event_results.json` is not uniformly Evaluator J data: Llama recovery rows and 18 Qwen2.5-14B recovery rows use a replacement `deepseek/deepseek-r1` evaluator. The fixed-evidence J/A comparison excludes these substituted readings and excludes Qwen2.5-14B entirely.

Four known failed rereads survive as stale copies of previous values: Qwen2.5-7B’s T3/C, C9/A and C13/C in repeat 3, and Qwen2.5-14B’s T1/C in repeat 2. The analysis excludes exactly these file–unit pairs. All 2,070 stored campaign probability vectors pass finite-value, range, normalization and option/outcome consistency checks. This validates record structure; it does not certify execution integrity or independently verify the historical event outcomes.

IV. MEASUREMENT AND ANALYSIS

A. Estimand and scoring

Let T_{mec} denote the stored evidence for campaign m , event e and condition c . An extraction configuration j produces $\hat{\mathbf{p}}_{mecj} = g_j(T_{mec}, q_e, \mathcal{O}_e)$, where q_e is the question and $\mathcal{O}_e$ the outcome list. The extraction function includes the model, prompt, generation settings, retries, serving and post-processing. The final forecast is an output of this composite system.

We use the unnormalised multicategory Brier score:

$$BS(\mathbf{p}, y) = \sum_{o \in \mathcal{O}} (p_o - \mathbf{1}\{o = y\})^2, \quad 0 \leq BS \leq 2. \quad (1)$$

For a uniform vector this equals $1 - 1/K$. A uniform vector is a legitimate probabilistic forecast; it is not automatically an error or a missing answer. The term *near-uniform* is an operational output diagnostic:

$$F(\mathbf{p}) = \mathbf{1} \left\{ \max_o |p_o - 1/K| < 0.005 \right\}. \quad (2)$$

We also report sensitivity to this tolerance. Near-uniform units remain in all forecasting score summaries. Excluding them because they are uninformative would change the estimand and could selectively remove difficult cases.

Directional accuracy credits a tie among h maximal-probability options by $1/h$ if the outcome belongs to that tie, and zero otherwise. The mean $1/K$ is reported as a descriptive uniform-choice reference, not a competitive forecasting baseline and not a null model for substantive event predictability.

B. Repeated readings and aggregation

For Evaluator A, the principal descriptive forecast uses the per-unit mean of all available probability vectors. Brier is then computed on that mean. A sensitivity analysis

instead averages the Brier scores of individual reads. These are different quantities:

$$\frac{1}{J} \sum_j BS(\mathbf{p}_j, y) = BS(\bar{\mathbf{p}}, y) + \frac{1}{J} \sum_j \|\mathbf{p}_j - \bar{\mathbf{p}}\|^2. \quad (3)$$

Averaging vectors therefore changes the forecast and can improve its score through ensembling. It does not simply reveal an otherwise hidden true swarm forecast. Evaluator C has only one read, so comparisons against the Evaluator A ensemble also differ in measurement budget.

Within-unit Brier range and exact vector agreement describe repeatability. Exact agreement is a strict diagnostic; disagreement can be numerically small, and range tends to increase with the number of reads. These quantities should not be used as calibrated thresholds for which population effects are measurable. Nor does inter-evaluator correlation establish agreement or validity.

C. Dependence and uncertainty

The four campaigns answer the same events. We first form each event’s mean paired contrast across the four fixed campaigns, then resample events with replacement. All observations belonging to a sampled event are kept together. We use 100,000 percentile bootstrap replicates, NumPy’s generator with seed 7, and 2.5th/97.5th percentiles. The estimand is the equal-event average over these four campaigns, not an average over a population of independently sampled models. Intervals condition on the observed simulations and evaluator readings.

The 20-event subset retains the prior analysis’s event selection to permit comparison. The selection predates this revision but is not preregistered. We call it the *post-release subset*, never contamination-clean: its outcomes resolve after the stated campaign-model release dates, but no corresponding protection exists for the later evaluator deployments or for retrospectively authored inputs.

Event bootstrap still assumes independent event clusters. C1, C5 and S2–S5 share the 2024 US election. We therefore additionally group these six events into one cluster (15 clusters for 20 events), and separately omit them (14 events). Grouped resampling preserves all events in a selected cluster and computes their event-weighted mean. These sensitivity choices are retrospective and do not exhaust every possible dependency in the catalogue.

We do not report the previous pooled Poisson-binomial p -value. It treated repeated model–event records as independent and rounded a fractional tie-credit total to an integer threshold. Neither operation supplies the exact sampling distribution of the reported accuracy statistic. More generally, no analysis in this paper is offered as a multiplicity-adjusted confirmatory test. Intervals across models, evaluators, variants and exclusions describe sensitivity rather than a search for one significant result.

V. FIXED-EVIDENCE EXTRACTION SENSITIVITY

Table I reports the primary audit. Mistral and Qwen2.5-7B each contribute all 60 simulated B/C units. Llama

contributes 12 units from its six unaffected events. The analysis verifies equality of stored evidence text, question, option list and ground truth for each pair, and aborts on mismatch.

TABLE I
NEAR-UNIFORM READINGS ON MATCHED SIMULATED EVIDENCE.
COUNTS REFER TO THE SAME UNITS WITHIN EACH ROW.

Campaign	Units	Evaluator J	Evaluator A
Llama-3.1-8B	12	12	0
Mistral-7B	60	19	1
Qwen2.5-7B	60	0	0
Total	132	31	1

The change from 31/132 to 1/132 establishes that the reported probability vectors are not invariant to extraction configuration. In the unaffected Llama subset all 12 original readings are near-uniform and none of their replacements are. This subset was selected by execution integrity, not randomly; it establishes that a large difference can occur, not its prevalence in arbitrary simulations.

The counts are unchanged using numerical exact-uniformity (maximum deviation at most 10^{-12}) or tolerance 0.01; at tolerance 0.02 the J count becomes 32 while the A count stays one. The contrast is therefore not generated by the particular 0.005 threshold in this paired material.

This is a fixed-*evidence* comparison, not a byte-identical HTTP-request experiment. Model, serving, generation budget, schema handling and possibly prompt/validation details differ across the historical configurations. The event-specific micro-question catalogue also underwent correction. The comparison locates sensitivity in the extraction stage as deployed; it does not isolate evaluator architecture from those other factors.

Most importantly, a non-uniform replacement is not automatically a more faithful reading. No human-labelled reference distribution or known-consensus synthetic transcript set was used to decide which extractor represents the discussion correctly. The replacement may recover information omitted by the first extractor, import outside knowledge, or become overconfident. The present records cannot distinguish these explanations.

A. Repeatability within the replacement evaluator

TABLE II
EVALUATOR A REPEATABILITY ACROSS 90 UNITS PER CAMPAIGN.
READS ARE AVAILABLE FILES; FOUR STALE FILE-UNIT READINGS ARE EXCLUDED. EXACT AGREEMENT CONCERNS PROBABILITY VECTORS, NOT ROUNDED BRIER VALUES.

Campaign	Reads	Exact	Mean BS range
Llama-3.1-8B	4	3/90	0.330
Mistral-7B	4	0/90	0.375
Qwen2.5-7B	4	4/90	0.280
Qwen2.5-14B	3	3/90	0.208

Table II shows substantial variability in repeated extraction. Mean within-unit Brier ranges are 0.208–0.375. Repeated reads here are repeated requests through the recorded deployment, not independent evaluators and not new simulations. The archive does not establish whether stochastic decoding, retries, backend nondeterminism or provider routing generated each difference. Because some requests failed and their stale copies are excluded, read counts are not perfectly balanced even within a campaign. The ranges support reporting repeatability alongside forecast quality; they do not provide a universal noise floor or an optimal number of required reads.

VI. SOURCE AUDIT AND EXISTING MECHANISM PROBES

A. What the code establishes

The archived client includes a keyword-based bypass for model identifiers containing strings such as `r1` or `reasoning`, and a separate local-quantization bypass. The identifier `deepseek-r1-14b` matches the reasoning filter. This documents a path that disables constraints for the configured name. Complete historical request logs are unavailable, so actual constraint status cannot be independently verified for every July request.

Schema enforcement and semantic correctness are different properties. JSON-object mode guarantees neither the nested schema nor meaningful probabilities. A schema-valid vector can be uniform, incorrect or unrelated to the transcript. Evaluator A used JSON-object handling; Evaluator C’s strict-schema requests were documented as falling back after schema rejection. Neither is a verified, uniformly strict schema-enforced reference instrument.

Silent defaults complicate diagnosis. Historical validation could accept incomplete micro-question mappings and fill them with missing-value labels. A deployed normalization path could also replace non-positive matched mass with a uniform vector. The final probabilities do not consistently preserve the raw completion and retry history needed to assign the origin of every uniform vector. The observations cannot establish that uniformity was a deliberate hedge by the model.

B. Cross-evaluator transfer

An existing follow-up sent Mistral’s 60 simulated evidence records to five additional unconstrained deployments. The stored experiment reports 8 near-uniform outputs among 58 parsed Qwen3-14B responses; six of these eight occur among the 19 units that Evaluator J flattened. Other deployments report zero near-uniform outputs among 56 DeepSeek-R1, 54 DeepSeek-R1-Distill-Llama-70B, 57 Qwen3-32B and 60 Ministral-14B responses.

These are counts among usable responses, not success rates over all attempted calls. Missing responses cannot be assumed missing at random. The simplified transfer prompt differs from the production rubric; an additional

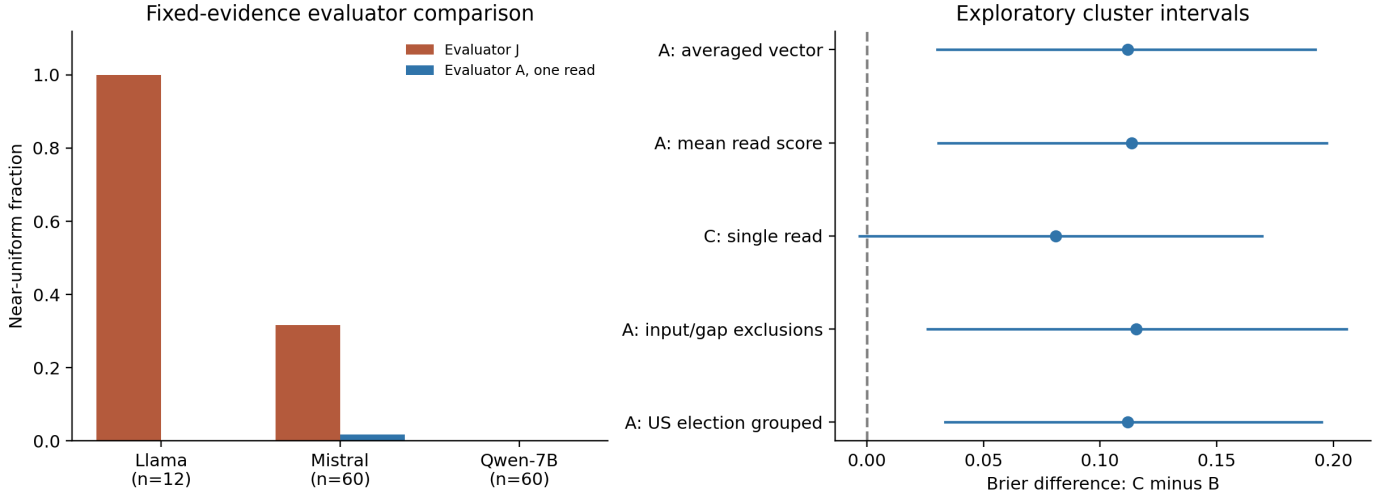


Fig. 1. Left: fixed-evidence near-uniform counts expressed as fractions (Table I). Right: exploratory Brier contrasts under alternative extraction and clustering choices (Table V). Bars on the left are observed fractions, not independent-agent estimates. Intervals on the right account for repeated campaigns within events; the grouped row additionally joins six US-election events.

DeepSeek-R1 production-prompt control reports zero near-uniform outputs among 60 scored units. This control does not make every cross-model comparison configuration-matched. The overlap identifies inputs on which two deployments return the same diagnostic pattern. It does not establish a model-size boundary: size, training and serving are not independently varied.

C. Truncation and requested enforcement

The existing grid varies four evidence lengths and whether output constraints are requested for a hosted Qwen3-14B proxy. Truncation retains parts of the log and subsamples representative actions; it changes content and speaker coverage together with character count. The original Evaluator J deployment was not available for this experiment.

TABLE III

QWEN3-14B GRID: NEAR-UNIFORM/USABLE READINGS OUT OF 60 ATTEMPTED UNITS PER CELL. ON DENOTES REQUESTED ENFORCEMENT; ACTUAL STRICT-SCHEMA ACCEPTANCE IS NOT VERIFIED FOR EVERY RESPONSE.

Length target	Off	On
2k	7/59	5/59
5k	4/60	2/60
8k	0/60	4/60
full	1/60	0/60

The unconstrained shortest arm has 7/59 near-uniform readings and the full arm 1/60. Pairing the 59 units with both responses yields six short-only near-uniform outcomes and zero full-only outcomes. Shortening did not reduce near-uniformity in this proxy experiment. A descriptive association between longer evidence and near-uniformity in the original Mistral/J observations cannot be promoted to a demonstrated length mechanism.

Across all lengths there are 238 matched off/on pairs because missing units differ between arms. We average paired indicator differences within each event and resample the 30 events, preserving conditions and length levels. The equal-event off-minus-on estimate is approximately $+0.005$, with interval $[-0.029, +0.042]$. The marginal totals, 12/239 and 11/239, should not be treated as independent binomial samples. This experiment does not detect a clear effect of requesting enforcement; it neither proves equivalence nor establishes that enforcement has no effect in the original deployment.

These probes bound candidate explanations without identifying the cause of Evaluator J’s outputs. The extraction-sensitivity observation does not require either a length mechanism or a schema mechanism to be true.

VII. EXPLORATORY FORECASTING RESULTS

A. Post-release subset

The 20 selected events yield 80 campaign–event records per simulation condition. Under Evaluator A’s averaged vector, B accuracy is 81.9%, comprising 65 outright correct modal outcomes and one two-way tie credited by one half. The mean uniform-choice reference is 32.6%. These are descriptive statistics of retrospective pipeline outputs.

TABLE IV

POST-RELEASE SUBSET, 20 EVENTS PER CAMPAIGN. ACCURACY USES TIE CREDIT; BRIER USES EVALUATOR A’S AVERAGED VECTOR. LABELS DO NOT IMPLY MATCHED SERVING.

Campaign	Acc. B	Acc. C	BS B	BS C
Llama-3.1-8B	0.675	0.500	0.545	0.613
Mistral-7B	0.850	0.650	0.378	0.441
Qwen2.5-7B	0.850	0.550	0.348	0.520
Qwen2.5-14B	0.900	0.800	0.289	0.432

The contrast $BS_C - BS_B$ is positive when B scores better. Table V presents all sensitivity summaries selected for this retrospective reanalysis; none is a preregistered hypothesis test.

The averaged-vector and mean-read-score summaries under A are similar here, so the B/C estimate is not solely a vector-ensembling artifact. Excluding the input/gap events retains a positive estimate. Grouping or omitting six US-election events also retains a positive estimate under A. These checks do not remove every shared cause of event outcomes.

Evaluator C gives $+0.081[-0.003, +0.170]$, whose interval includes zero. Its single-read budget differs from A’s repeated-read budget. This is limited robustness across the available measurement choices, not a clean test that one evaluator detects an effect and another does not. A statistical difference between the two effects is not established.

B. What the contrast cannot identify

The post-release filter concerns the campaign models, not all LLMs in the pipeline. Evaluators deployed in 2026 may know the outcomes of events from 2024–2025. The probability prompt exposes identifiable questions, and instructions to use only the discussion do not demonstrate compliance. Dossiers and perturbations were assembled retrospectively; timestamps, source fidelity and freedom from hindsight were not independently established for every claim. Stripping URLs or softening deterministic language cannot remove knowledge already encoded in a model.

The accuracy figures therefore do not establish leakage-free forecasting. They may reflect genuine evidence use, pre-existing model knowledge, evaluator knowledge, retrospective input construction, or a combination. Ground-truth labels are used as recorded; this study does not independently adjudicate every market resolution or the mutual exclusivity of every curated option set.

The B/C contrast describes sensitivity of the complete simulated-and-extracted pipeline to its input conditions. It cannot attribute that sensitivity specifically to agent-to-agent deliberation. A direct call, an extractor reacting to a changed transcript, or different injected wording may also produce a B/C difference. Relevance is not isolated from length, and C6/T6/T8 do not implement the intended relevance manipulation.

VIII. EXISTING SINGLE-AGENT BASELINE

The post-campaign baseline contains 240 calls: 20 events, four hosted model arms and three calls per event/arm. Each receives the seed dossier and nominal B update, not the simulation transcript. The first call uses temperature zero, the other two temperature 0.7; these are not identically distributed replicates.

The principal summary averages individual-call Brier scores. It estimates the mean score of the observed three-setting mixture, not a fixed-temperature single-call policy.

The script also reports the first call alone and the score of the three-vector ensemble. No added retrieval step was executed: the dossier and injection were supplied directly in the prompt.

Only Llama-3.1-8B and Qwen2.5-7B have nominal model identities matching their campaigns; identical served weights are not independently verified, and serving and precision are not fully matched. Ministral-8B and Qwen3-14B differ from the Mistral-7B and Qwen2.5-14B campaigns and are secondary comparisons.

Pooling the identity-matched arms by event gives $BS_{CoT} - BS_{swarm} = -0.106[-0.226, +0.008]$. The point estimate favours CoT and the interval includes zero. This is no demonstrated swarm advantage in the available comparison, not proof of equivalence or non-inferiority. Some individual or ensemble variants exclude zero in CoT’s favour; these are exploratory and do not change that scope.

Across all four, partly unmatched arms, individual-call accuracy is 83.4%, compared with 81.9% for the swarm under A’s averaged vectors. This does not establish equal capability: model identity, serving and aggregation are unbalanced. There is no single-agent C arm. A factorial comparison of single-agent versus simulation crossed with the same B/C inputs would be needed to estimate additional content sensitivity from deliberation. The present data do not identify that interaction.

IX. IMPLICATIONS AND LIMITS

A. A practical measurement protocol

A reproducible evaluation should preserve the event question, ordered options, evidence bytes, full prompt, model identifier, serving revision when available, generation parameters, requested and accepted output format, raw completion, retry history and probability post-processing. A final vector and an error flag are insufficient to diagnose the defects found here.

Repeated extraction and a second evaluator are useful diagnostics, not substitutes for external validity. Read counts should follow a precision/cost analysis; this study does not justify a universal minimum of three reads. A labelled synthetic transcript or blinded human reference can test extraction fidelity. A question-only baseline can measure what a later evaluator knows without the simulation. These are recommendations, not experiments executed in this revision. Repairing a schema bypass is sound engineering even when its effect on near-uniformity is unknown. Valid JSON, a non-uniform forecast and a faithful interpretation are separate requirements.

B. Limits of the evidence

The primary comparison is conditional on a selected artifact set and a historical deployment. The original endpoint’s disappearance prevents exact intervention on that deployment. An open-weight model identifier exists,

TABLE V

EXPLORATORY SENSITIVITY OF $BS_C - BS_B$. INTERVALS RESAMPLE EVENTS, EXCEPT THE US-ELECTION-GROUPED ROW (15 CLUSTERS OVER 20 EVENTS). INPUT/GAP EXCLUSION REMOVES C6, T6, T8 AND C9 FROM ALL CAMPAIGNS. EXCLUSIONS WERE CHOSEN AFTER INSPECTION OF THE AUDIT TRAIL; INTERVALS DO NOT CORRECT FOR THIS SELECTION.

Analysis	Events	Mean [95% interval]
Evaluator A, averaged vector	20	+0.112[+0.030, +0.192]
Evaluator A, mean read score	20	+0.114[+0.031, +0.197]
Evaluator C, single read	20	+0.081[−0.003, +0.170]
A, excluding input/gap events	16	+0.116[+0.026, +0.206]
A, US election grouped	20	+0.112[+0.034, +0.195]
A, excluding US election	14	+0.117[+0.013, +0.218]

TABLE VI

EXISTING CoT BASELINE. POSITIVE $BS_{\text{CoT}} - BS_{\text{swarm}}$ FAVOURS THE SWARM. IDENTITY MATCHING DOES NOT IMPLY SERVING/PRECISION MATCHING. PER-ARM INTERVALS ARE EXPLORATORY EVENT BOOTSTRAPS.

CoT arm	Identity matched?	Accuracy	Brier difference [95% interval]
Llama-3.1-8B	Yes	0.778	−0.181[−0.366, −0.020]
Qwen2.5-7B	Yes	0.790	−0.030[−0.134, +0.082]
Mistral-8B	No	0.912	−0.170[−0.371, +0.010]
Qwen3-14B	No	0.858	−0.008[−0.114, +0.092]

but restoring weights under a new stack would be a new experiment, not a proven replay.

Repeated reads condition on one realised simulation per condition and cannot estimate simulation-run variability. Nominally 300 agents sharing a model do not supply 300 independent replications. The 30-event convenience catalogue supports limited generalization to other events, languages, forecasting horizons and deployments.

The seven-dimension rubric, convergence telemetry and persona/action audits remain execution diagnostics, not validated measures of psychological or collective-reasoning constructs. Incomplete telemetry, placeholders and serving failures prevent the original convergence and optimal-configuration claims from being retained as results. Isolated failures of small deployments do not establish a general parameter-count exclusion rule.

No causal mechanism is rescued by an insignificant comparison, and an inability to reject a null is not proof of equality. The main finding is sensitivity of measured probabilities to extraction configuration; forecasting comparisons make its consequences and limits visible.

X. CONCLUSION

On 132 matched simulated evidence records, the original and replacement extraction configurations produce 31 and one near-uniform forecasts. Repeated reads of the replacement also vary substantially. These observations warrant treating an LLM extractor as part of the forecasting system.

The source audit and proxy experiments do not identify why the original configuration produced these vectors or which reading is faithful to the swarm. Forecasting contrasts remain retrospective, evaluator-dependent and affected by imperfect inputs. An existing single-agent baseline does not demonstrate a deliberation benefit. The

contribution is a reproducible audit of extraction sensitivity and the design boundaries it exposes, rather than a validated advantage or failure of collective reasoning.

APPENDIX A

REPRODUCTION AND ARTIFACT CONTRACT

The accompanying archive contains the benchmark definition in `01_benchmark`, scored records in `02_campaigns`, available traces in `03_traces`, existing follow-up observations in `04_experiments`, revised analysis in `05_analysis`, historical execution code in `07_pipeline` and superseded artifacts in `09_audit`. From the archive root:

```
python 05_analysis/scripts/paper_analysis.py
-data-root . -output-dir 08_paper
```

The script reads frozen observations and generates `paper_results.json`, `paper_tables.tex` and `fig_paper_audit.png`, without network calls. The JSON records settings and SHA-256 input hashes; tables consume generated values directly. A numerical validation suite checks estimands and data-contract failure cases.

The paired comparison validates evidence/question/options equality, not complete historical request equality. Four stale rereads are excluded by file-unit keys. Forecast metrics are recomputed from vectors rather than inherited fields. Historical scripts and reports are retained for provenance; they are not authoritative for current paper tables.

APPENDIX B

CORRECTION RECORD

The prior manuscript is retained under `09_audit/manuscript_20260826.tex` and is superseded. This revision replaces a demonstrated-mechanism claim with a fixed-evidence configuration observation; replaces contamination-clean language with post-release subset;

withdraws the independent-unit pooled accuracy test; uses event-cluster intervals; adds evaluator, aggregation, input and dependency sensitivities; corrects the B/C description; retains uniform vectors in scoring; and removes unvalidated convergence, size-threshold and optimal-configuration claims. No new model observations were collected. The defects remain available to audit and remain explicit limits on inference.

DATA AVAILABILITY

The replication package accompanies this manuscript and supports offline reanalysis. A permanent repository identifier has not yet been assigned in this working release; no public deposition or reviewer access is asserted. Scored records and evidence fields survive even where raw simulation traces were lost. Qwen2.5-14B’s original raw traces are unavailable for 21 events following scratch-storage deletion; recovery traces and the surviving original fragment are retained. The original evaluator deployment cannot be replayed exactly from the recorded artifacts. Reproducing the reported analysis and rerunning the historical system are distinct capabilities.

RESEARCH SCOPE

Inputs concern public events and were assembled for retrospective computational study. No new human-participant data were collected by the reanalysis. Synthetic personas and behaviours are not validated models of real populations. Results should not be interpreted as operational forecasting performance or as evidence about the beliefs of actual people.

REFERENCES

- [1] T. Gneiting and A. E. Raftery, *Strictly proper scoring rules, prediction, and estimation*, Journal of the American Statistical Association, vol. 102, no. 477, pp. 359–378, 2007, doi:10.1198/016214506000001437.
- [2] L. Zheng et al., *Judging LLM-as-a-judge with MT-Bench and Chatbot Arena*, Advances in Neural Information Processing Systems, 2023. <https://arxiv.org/abs/2306.05685>.
- [3] R. Haldar and J. Hockenmaier, *Rating roulette: Self-inconsistency in LLM-as-a-judge frameworks*, Findings of EMNLP, pp. 24986–25004, 2025, doi:10.18653/v1/2025.findings-emnlp.1361.
- [4] Z. Yang et al., *OASIS: Open agent social interaction simulations with one million agents*, arXiv:2411.11581, 2024. <https://arxiv.org/abs/2411.11581>.
- [5] E. Karger et al., *ForecastBench: A dynamic benchmark of AI forecasting capabilities*, arXiv:2409.19839, 2024. <https://arxiv.org/abs/2409.19839>.